Fiscal Year:	FY 2006	**Task Last Updated:**	FY 05/24/2007
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Project Title:	Adaptive Control of Alloy Microstructures		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Metals and alloys		
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Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
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Space Biology Special Category:	None		
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Zip Code:	02115-5005	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
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No. of Bachelor's Candidates:		**Monitoring Center:**	NASA MSFC
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Flight Program:			
Flight Assignment:	NOTE: Received no-cost extension to 9/30/2007 per PI (3/07)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Losert, Wolfgang (University of Maryland)		
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	Material microstructure is crucial in diverse areas of materials research from metallurgy to studies of biomaterials. In particular, the microstructure (structure on sub-micron to millimeter scales) often determines material strength and other key properties. Our research aims to develop a new approach toward understanding and controlling microstructural patterns. We propose a unique combination of quantitative phase-field modeling and novel perturbation experiments. The project will focus on microstructures of great technological relevance, cellular and dendritic arrays that develop during binary alloy directional solidification, though the basic approach to microstructure control should be more broadly applicable to a wide range of materials. In traditional directional solidification experiments, control of the microstructure is limited to global control through variations of the alloy composition and the growth conditions (externally imposed growth rate and temperature gradient). The evolution of the solid-liquid interface can only be influenced through		

Task Description:	changes of these parameters. We propose to pursue a new control approach where the interface evolution can be altered by spatially localized perturbations that can be applied at arbitrary times. These perturbations are chosen adaptively, i.e. based on the current state of the system, so as to guide the interface evolution into desired microstructures with optimal properties that would not otherwise be accessible. This approach exploits the extreme sensitivity of dendritic patterns to very small amplitude perturbations, of magnitude comparable to thermal fluctuations, and the fact that a wide range of structures can form under the same growth conditions due to the nonequilibrium nature of the solidification process. This research relies on new state-of-the-art experimental and computational tools: 1) A holographic laser tweezer array will be used to apply complex perturbations to directionally solidified impure succinonitrile crystals using a laser light field that can be shaped adaptively in response to the observed microstructure. Real time observations of the microstructure in three-dimensional bulk samples will be made possible by a two-photon confocal microscope, while observations in thin samples will utilize a standard optical microscope set-up. These techniques will allow us to create perturbations of the solid-liquid interface of arbitrary shape and wavelength (i.e. arbitrary spatial arrangements of UV light spots) as well as to locally perturb or remelt a desired subset of cell/dendrite tips in an array. 2) Simulations of a new alloy phase-field formulation (Karma, 2002b) will be used to model quantitatively the solid-liquid interface evolution in two and three dimensions with realistic materials parameters and to guide the design of the experiments. In a past collaboration, the PI and Co-I have already demonstrated that perturbative experiments and phase-field simulations can be combined successfully to obtain new insights into microstructural evolution beyond the beaten path of conventional approaches (Losert et al., 1998c). The use of the aforementioned new techniques will greatly enlarge the scope of this work by extending it to three dimensions and by making the comparison between simulations and experiments quantitative. Our dual objective is to use these tools to shed new light on long standing microstructural pattern formation questions that are of both fundamental and practical importance and to produce new microstructures with desired properties. We will focus our attention specifically on two closely related problems: the dynamical selection of the primary cell/dendrite array spacing, which is influenced by both the complex history of the solid-liquid interface evolution and the sidebranching dynamics, and the cell to dendrite transition in both two and three dimensions. In addition, we will investigate how adaptive control can be used to stabilize unstable microstructures using the planar interface as a first concrete t [task description truncated]
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	We have successfully developed and implemented a feedback control scheme to stabilize unstable cellular/dendritic patterns during the directional solidification of a binary alloy, which is one of the main goal of our project. The scheme is based on local heating of cell tips which protrude ahead of the mean position of all tips in the array. The feasibility of this scheme was demonstrated using phase-field simulations and, experimentally, using a real-time image processing algorithm, to track cell tips, coupled with a movable laser spot array device to heat the tips locally. Furhtermore, as predicted by the numerical calculations, cellular/dendritic arrays become stable with uniform spacing through the feedback control which is maintained with minimal heating. A key feature of this novel feedback control method is that it makes it possible to stabilize cellular/dendritic arrays with spacings well below the threshold for a well-known spatial period-doubling instability that eliminates one out of every two cells/dendrites along the array. More specifically, the feedback control scheme consists of injecting an amount of heat in a spotlike region centered at a cell/dendrite tip that is constant regardless of how far the targeted cell grows beyond the average cell/dendrite position. For a given cell, local heating at the tip is applied when the distance from the average cell position is larger than a predefined cutoff. Unlike other feedback stabilization schemes developed previously for planar fronts slightly above the critical velocity for the onset of morphological instability, our approach stabilizes efficiently highly unstable states and can be implemented experimentally with only a discrete number of controllable heating points. The use of other localized physical perturbations generated, e.g., by x-rays or ultrasound , could potentially make it possible to extend this control scheme to metallic alloys where the growth of much finer array structures with superior mechanical properties is of considerable practical interest. In addition, the ability to access experimentally unstable steady-state patterns provides an important new tool to enrich our fundamental understanding of pattern formation in directional solidification.
Bibliography Type:	Description: (Last Updated: 05/10/2026)
Articles in Other Journals or Periodicals	Pons AJ, Karma A, Akamatsu S, Newey M, Pomerance A, Singer H, Losert W. "Feedback control of unstable cellular solidification fronts." Physical Review E, in press, January 2007. , Jan-2007
Articles in Peer-reviewed Journals	Akamatsu S, Lee K, Losert W. "Control of eutectic solidification microstructures through laser spot perturbations." Journal of Crystal Growth 2006 Mar 15; 289(1): 331-8. http://dx.doi.org/10.1016/j.jcrysgro.2005.11.003 , Mar-2006
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